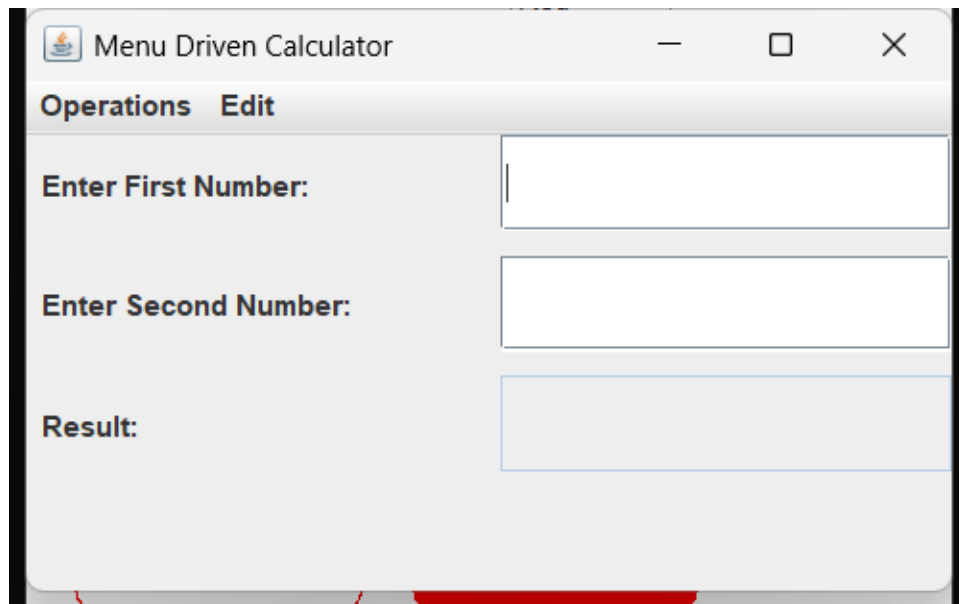


MENU DRIVEN CALCULATOR

PROGRAM OUTPUT SCREENSHOT



SOURCE CODE

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class MenuDrivenCalculator extends JFrame implements ActionListener {
    private final JTextField num1Field;
    private final JTextField num2Field;
    private final JTextField resultField;

    private final JMenuItem addVal, subVal, mulVal, divVal, clearVal, exitVal;

    public MenuDrivenCalculator() {
        setTitle("Menu Driven Calculator");
        setSize(400, 250);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new GridLayout(4, 2, 10, 10));

        num1Field = new JTextField();
        num2Field = new JTextField();
        resultField = new JTextField();
        resultField.setEditable(false);

        add(new JLabel("  Enter First Number:"));
        add(num1Field);
        add(new JLabel("  Enter Second Number:"));
        add(num2Field);
        add(new JLabel("  Result:"));
        add(resultField);

        JMenuBar menuBar = new JMenuBar();

        JMenu calcMenu = new JMenu("Operations");
        addVal = new JMenuItem("Add");
        subVal = new JMenuItem("Subtract");
        mulVal = new JMenuItem("Multiply");
        divVal = new JMenuItem("Divide");
        calcMenu.add(addVal);
        calcMenu.add(subVal);
        calcMenu.add(mulVal);
        calcMenu.add(divVal);

        JMenu editMenu = new JMenu("Edit");
        clearVal = new JMenuItem("Clear");
        exitVal = new JMenuItem("Exit");
        editMenu.add(clearVal);
        editMenu.addSeparator();
        editMenu.add(exitVal);

        menuBar.add(calcMenu);
        menuBar.add(editMenu);
        setJMenuBar(menuBar);

        addVal.addActionListener(this);
        subVal.addActionListener(this);
        mulVal.addActionListener(this);
        divVal.addActionListener(this);
        clearVal.addActionListener(this);
        exitVal.addActionListener(this);
    }
}
```

```

        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        Object source = e.getSource();

        if (source == exitVal) {
            System.exit(0);
        } else if (source == clearVal) {
            num1Field.setText("");
            num2Field.setText("");
            resultField.setText("");
            return;
        }

        try {
            double n1 = Double.parseDouble(num1Field.getText().trim());
            double n2 = Double.parseDouble(num2Field.getText().trim());
            double res = 0;

            if (source == addVal) {
                res = n1 + n2;
            } else if (source == subVal) {
                res = n1 - n2;
            } else if (source == mulVal) {
                res = n1 * n2;
            } else if (source == divVal) {
                if (n2 == 0) {
                    JOptionPane.showMessageDialog(this, "Division by zero is not allowed.",
"Math Error", JOptionPane.ERROR_MESSAGE);
                    return;
                }
                res = n1 / n2;
            }
            resultField.setText(String.valueOf(res));
        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(this, "Please enter valid numeric inputs.",
"Input Error", JOptionPane.ERROR_MESSAGE);
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(new MenuDrivenCalculator());
    }
}

```